

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

5 - Linear State Feedback with an Observer

© Prof. Victor Solo

School of Electrical Engineering and
Telecommunications

UNSW

Sydney

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Topics

- 1 Observers (Virtual Sensors)
- 2 Linear State Feedback+ Observer = LSFBO
- 3 Transfer Function View of LSFBO
- 4 LSFBO Design by polynomial methods

References

Goodwin et al, chapters 17,18.

Kailath, chapters 3,4

OBSERVERS

- State feedback

requires all states be sensed. If only $u(t)$, $y(t)$ are sensed (2 dof) we are led to try to estimate the state.

- Open-loop estimation

We know A, b, c , we measure $u(t)$, $y(t)$ $t \geq 0$.

And if c, A is observable, we can recover x_0 . So simulate $\dot{x} = Ax + bu$ with the recovered x_0 and feed in the measured $u(t)$.

But if the system is unstable the simulation blows up. Otherwise noise sensitivity is a problem.

So we are led to a feedback configuration.

Observer via Feedback

We need an error signal to feedback. If $\hat{x}(t)$ is the generated state estimator we cannot get the state error $\tilde{x}(t) = x(t) - \hat{x}(t)$ but we can get

$$\begin{aligned} e(t) &= y(t) - \hat{y}(t) \\ &= y(t) - c^T \hat{x}(t) \end{aligned}$$

So we feed this back yielding

$$\dot{\hat{x}} = A\hat{x} + bu + l(y - c^T \hat{x})$$

where l is the observer gain to be chosen.

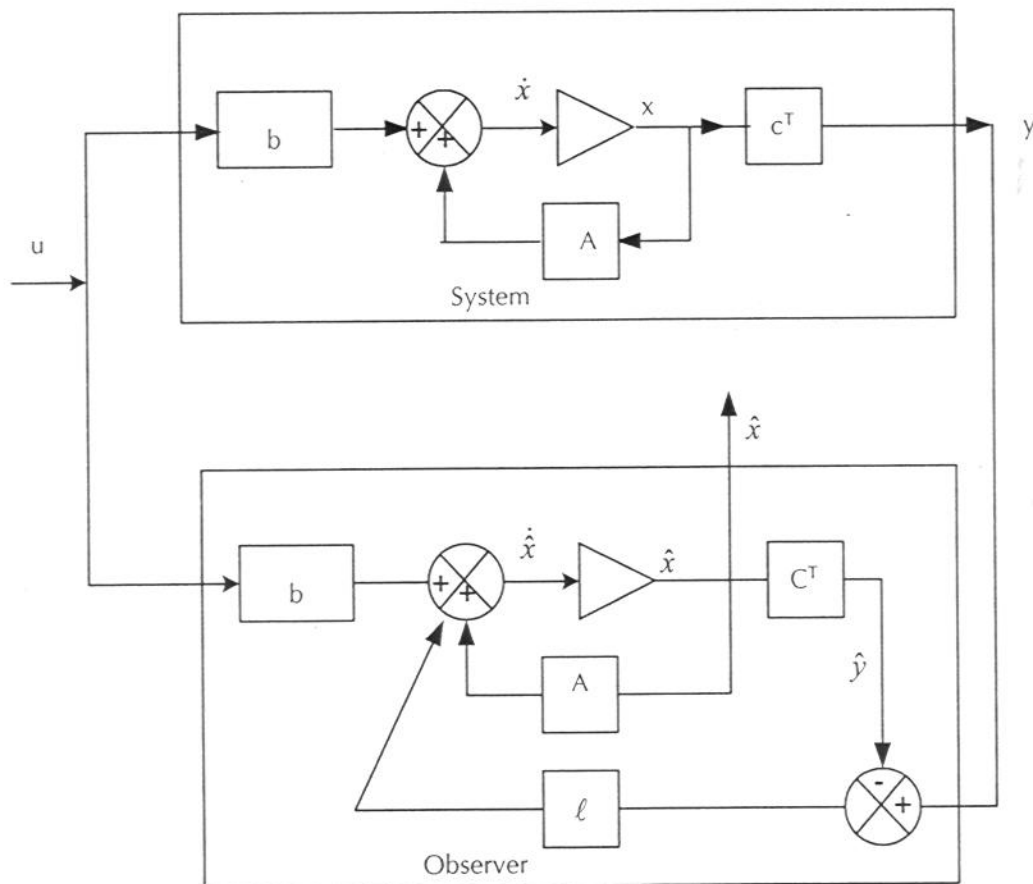


Fig. 5-1: Observer

OBSERVER DESIGN

The state error obeys

$$\begin{aligned}\dot{\tilde{x}} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + bu - [A\hat{x} + bu + lc^T(x - \hat{x})] \\ &= A(x - \hat{x}) - lc^T(x - \hat{x}) \\ &= (A - lc^T)\tilde{x}\end{aligned}$$

Now we need to choose l to ensure this error system is stable.

So we specify a characteristic polynomial $a_l(s)$ with poles in desired stable locations.

OBSERVER DESIGN -Continued

And we must choose l to ensure

$$\left| sI - A + lc^T \right| = a_l(s)$$

But this equals $\left| sI - A^T + cl^T \right|$

And so we have the same problem as in SFB design (with $b \equiv c, k \equiv l$).

Solution is thus given by the Bass-Gura formula

$$\begin{aligned} l &= \mathcal{C}^{-T}(A^T, c) \mathcal{A}_-^{-1} g_d, \quad g_{di} = a_{l,i} - a_i \\ &= \mathcal{O}^{-1}(c, A) \mathcal{A}_-^{-1} g_d \\ &= [\mathcal{A}_- \mathcal{O}(c, A)]^{-1} g_d \end{aligned}$$

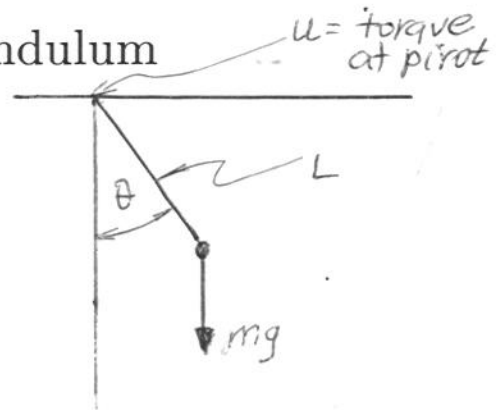
Clearly this requires c, A be observable. Again near lack of observability will force large observer gains which may be impractical.

OBSERVER - Example

State estimator for controlled pendulum

ODE, (for small θ)

$$\begin{aligned}\ddot{\theta} &= -\omega_0^2 \theta + u \\ \omega_0^2 &= \frac{g}{L}\end{aligned}$$



SS model

$$x_1, x_2 = \theta, \dot{\theta}, y = \theta$$

$$\begin{aligned}\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x\end{aligned}$$

Characteristic Polynomial is

$$\begin{aligned}a(s) &= |sI - A| = s^2 + \omega_0^2 = s^2 + 0s + \omega_0^2 \\ &= s^2 + a_1s + a_2\end{aligned}$$

[poles are at $\pm j\omega_0$]

Place observer poles at $3(-\omega_0)$ giving a closed loop characteristic equation

$$\begin{aligned}a_l(s) &= (s + 3\omega_0)^2 = s^2 + 6\omega_0s + 9\omega_0^2 \\ &= s^2 + a_{l,1}s + a_{l,2}\end{aligned}$$

OBSERVER - Example - Continued

Apply Bass - Gura $l = [\mathcal{A}_- \mathcal{O}(c, A)]^{-1} g_d$

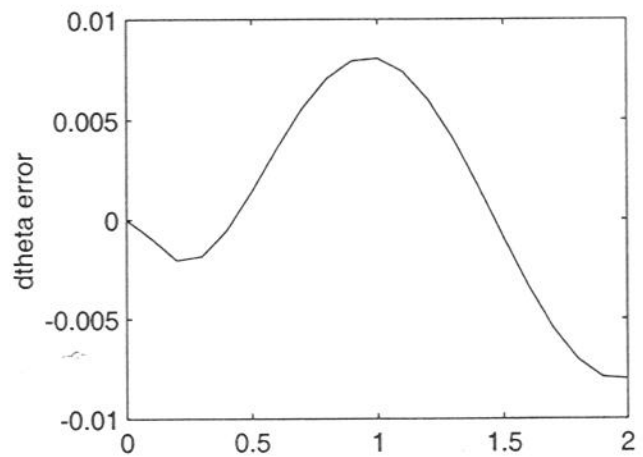
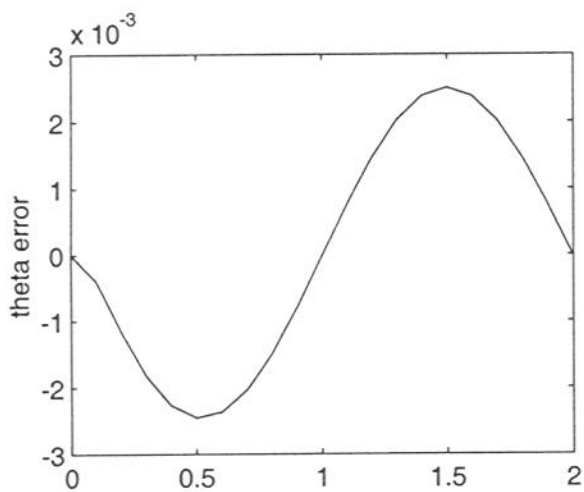
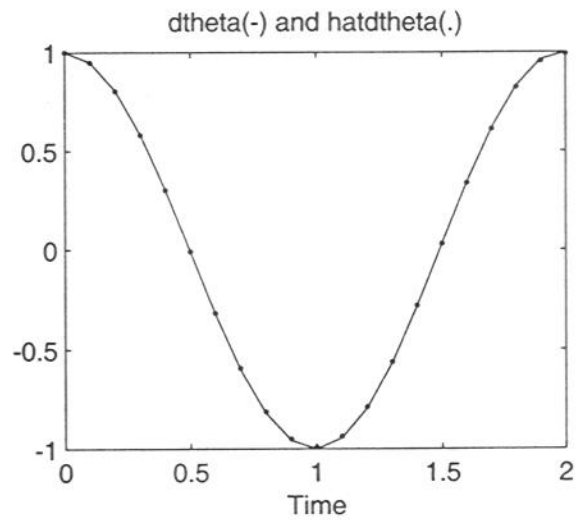
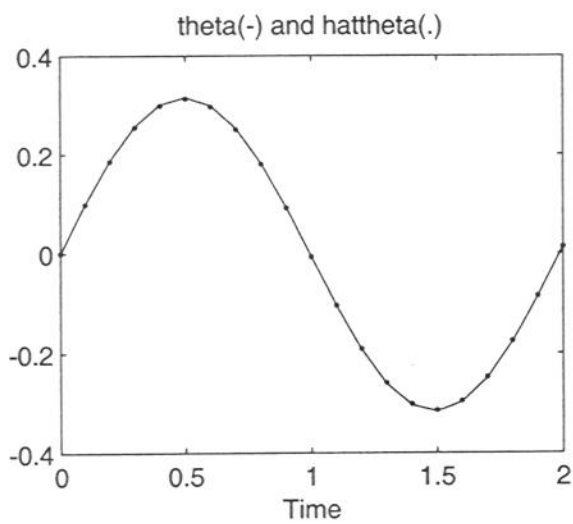
$$g_d = \begin{pmatrix} a_{1,l} - a_1 \\ a_{2,l} - a_2 \end{pmatrix} = \begin{pmatrix} 6\omega_0 - 0 \\ 9\omega_0^2 - \omega_0^2 \end{pmatrix} = \begin{pmatrix} 6\omega_0 \\ 8\omega_0^2 \end{pmatrix}$$

$$\mathcal{A}_- = \begin{pmatrix} 1 & 0 \\ a_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathcal{O}(c, A) = \begin{pmatrix} c^T \\ c^T A \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow l &= [\mathcal{A}_- \mathcal{O}(c, A)]^{-1} g_d \\ &= \begin{pmatrix} 6\omega_0 \\ 8\omega_0^2 \end{pmatrix} \end{aligned}$$

OBSERVER - Example - Continued II



COMBINED LSFB + OBSERVER

We put the two together in Fig.5.2

System

$$\dot{x} = Ax + bu, \quad y = c^T x$$

$$u = r - k^T \hat{x}$$

Observer

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + bu + l(y - c^T \hat{x}) \\ &= A\hat{x} + bu + lc^T(x - \hat{x})\end{aligned}$$

Joint analysis is simplified by using the state estimation error $\tilde{x} = x - \hat{x}$

$$\Rightarrow \dot{\tilde{x}} = A\tilde{x} - lc^T \tilde{x} = (A - lc^T)\tilde{x}$$

while the system equation becomes

$$\begin{aligned}\dot{x} &= Ax + br - bk^T \hat{x} \\ &= (A - bk^T)x + br - bk^T \tilde{x}\end{aligned}$$

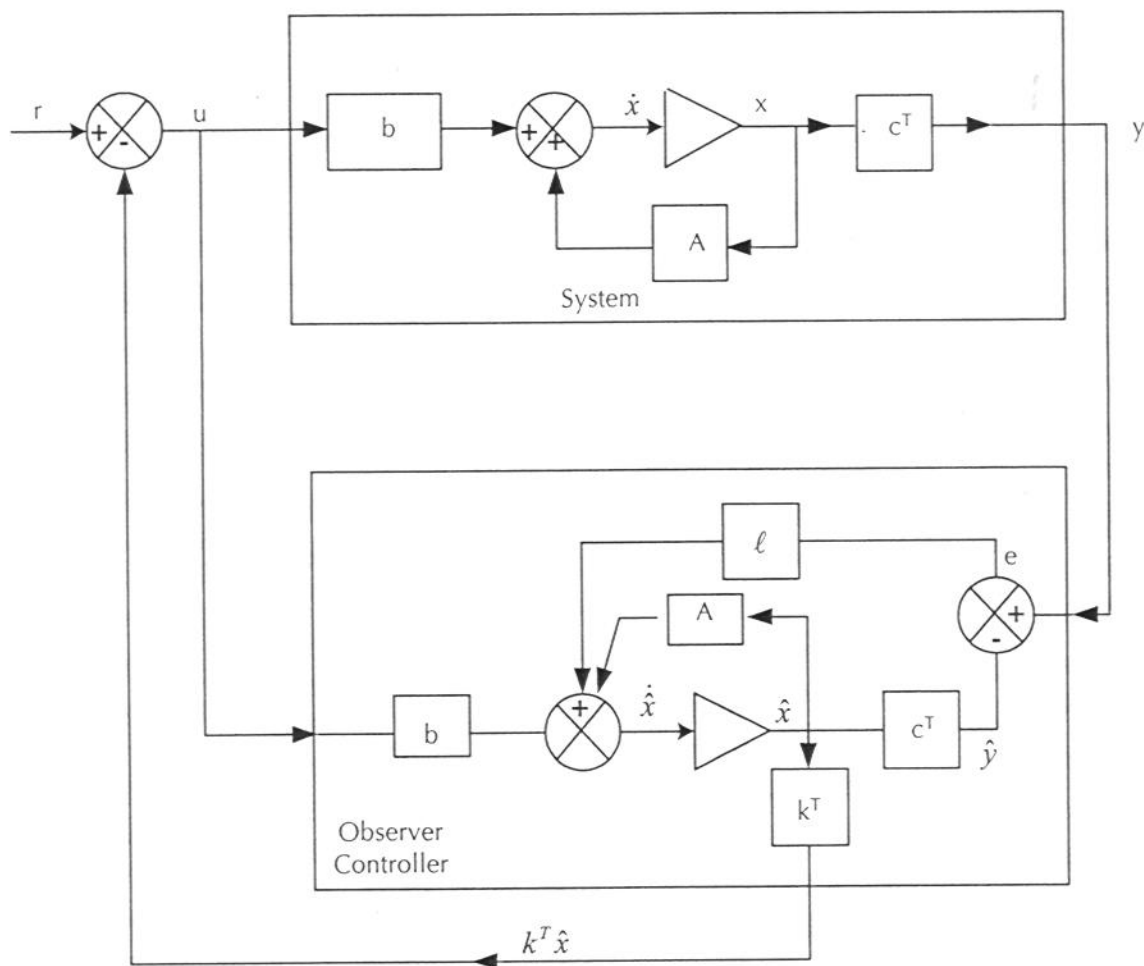


Fig. 5.2: LSFB + Observer

COMBINED LSFB + OBSERVER

-Continued I

Putting these together gives

$$\begin{pmatrix} \dot{x} \\ \dot{\tilde{x}} \end{pmatrix} = \begin{pmatrix} A - bk^T & -bk^T \\ 0 & A - lc^T \end{pmatrix} \begin{pmatrix} x \\ \tilde{x} \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix} r$$

Taking Laplace transforms and solving gives

$$\begin{aligned} & \begin{pmatrix} sI - A + bk^T & bk^T \\ 0 & sI - A + lc^T \end{pmatrix} \begin{pmatrix} \bar{x} \\ \bar{\tilde{x}} \end{pmatrix} \\ &= \begin{pmatrix} b \\ 0 \end{pmatrix} \bar{r} + IC \\ &\Rightarrow \begin{pmatrix} \bar{x} \\ \bar{\tilde{x}} \end{pmatrix} = \\ & \begin{pmatrix} \boxed{\boxed{}}_k^{-1} & \boxed{\boxed{}}_k^{-1} bk^T \boxed{\boxed{}}_l^{-1} \\ 0 & \boxed{\boxed{}}_l^{-1} \end{pmatrix} \left(\begin{pmatrix} x_0 \\ \tilde{x}_0 \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix} \bar{r} \right) \\ & \boxed{\boxed{}}_k = sI - A + bk^T, \quad \boxed{\boxed{}}_l = sI - A + lc^T \end{aligned}$$

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LSFB+O Continued II

For future reference recall the definitions

$$a_k(s) = \det(sI - A + bk^T)$$

$$a_l(s) = \det(sI - A + lc^T)$$

Also note

$$\begin{aligned} a_k(s) &= \det(sI - A + bk^T) \\ &= |sI - A| \left| I + (sI - A)^{-1}bk^T \right| \\ &= a(s)(1 + k^T(sI - A)^{-1}b) \\ &= a(s)\left(1 + \frac{\bar{k}(s)}{a(s)}\right) \\ &= a(s) + \bar{k}(s) \end{aligned}$$

Similarly $a_l(s) = a(s) + \bar{l}(s)$

LSFB + OBSERVER - Continued III

Continuing we get a closed loop system

$$\begin{aligned}\bar{y} &= c^T \bar{x} \\ &= c^T \begin{bmatrix} -1 \\ k \end{bmatrix} x_0 + c^T \begin{bmatrix} -1 \\ k \end{bmatrix} b k^T \begin{bmatrix} -1 \\ l \end{bmatrix} \tilde{x}_0 \\ &+ c^T \begin{bmatrix} -1 \\ k \end{bmatrix} b \bar{r} \\ &= \text{initial condition transients} + \text{forced response}\end{aligned}$$

We already showed

$$c^T \begin{bmatrix} -1 \\ k \end{bmatrix} b = \frac{b(s)}{a_k(s)} = CLTF$$

Note here that the CLTF does not depend on the observer (modes) But all modes do effect the initial conditions. We will see below however that observer modes do effect disturbance and noise transfer functions.

So even though we now estimate the state we get the same CLTF.

LSFB + OBSERVER - Example

Consider design of a 2DOF LSFBO to stabilise a double integrator.

- SS model

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$
$$b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

- Closed loop Behaviour. Suppose we specify

Closed Loop Poles

$$a_k(s) = s^2 + \sqrt{2}s + 1, \quad (\omega_n = 1, \zeta = .7)$$

Observer Poles

$$a_l(s) = s^2 + 5s + 25, \quad (\omega_n = 5, \zeta = .5)$$

- Characteristic polynomial

$$a(s) = |sI - A| = s^2 + 0.s + 0$$

LSFB + OBSERVER - Example Continued

- Control Gain Design

$$k = [\mathcal{A}_- \mathcal{C}^T(A, b)]^{-1} g_{dk}$$

$$g_{dk} = \begin{pmatrix} a_{k1} - a_1 \\ a_{k2} - a_2 \end{pmatrix} = \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix}$$

$$\mathcal{A}_- = \begin{pmatrix} 1 & 0 \\ a_1 & 1 \end{pmatrix} = I$$

$$\mathcal{C}(A, b) = (b, Ab) = I$$

$$\Rightarrow k = \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix}$$

LSFB + OBSERVER - Example Continued II

- Observer Gain Design

$$\begin{aligned}l &= [\mathcal{A}_- \mathcal{O}^T(c, A)]^{-1} g_{dl} \\g_{dl} &= \begin{pmatrix} a_{l1} - a_1 \\ a_{l2} - a_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 25 \end{pmatrix} \\ \mathcal{A}_- &= \begin{pmatrix} 1 & 0 \\ a_1 & 1 \end{pmatrix} = I \\ \mathcal{O}(c, A) &= \begin{pmatrix} c^T \\ c^T A \end{pmatrix} = I \\ \Rightarrow l &= \begin{pmatrix} 5 \\ 25 \end{pmatrix}\end{aligned}$$

TRANSFER FUNCTION VIEW

If we look at Fig.5.2 carefully we see that effectively u, y are being filtered to produce the feedback signal. So we have a 2 DOF feedback control system.

We thus expect to be able to find a transfer function view of this as in Fig.5.3

We look then for an expression of the form

$$k^T \hat{\bar{x}} = C_u(s)\bar{u} + C_y(s)\bar{y}$$

And we look for a polynomial method of finding $C_u(s), C_y(s)$.

TRANSFER FUNCTION VIEW - Continued I

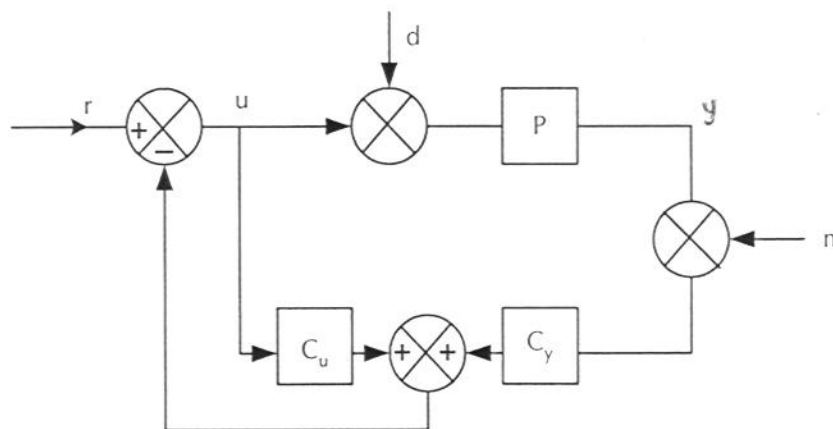


FIGURE 5.3: 2DOF Transfer function
view of LSFB + Observer

TRANSFER FUNCTION VIEW - Continued II

Looking at the observer we can reorganise it appropriately thus

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + bu + l(y - c^T \hat{x}) \\ &= (A - lc^T)\hat{x} + bu + ly\end{aligned}$$

Giving the laplace transform

$$\hat{\bar{x}} = (sI - A + lc^T)^{-1}b\bar{u} + (sI - A + lc^T)^{-1}l\bar{y} + IC$$

and so the control signal is given by

$$\begin{aligned}k^T \hat{\bar{x}} &= C_u(s)\bar{u} + C_y(s)\bar{y} \\ C_u(s) &= k^T (sI - A + lc^T)^{-1}b \\ C_y(s) &= k^T (sI - A + lc^T)^{-1}l\end{aligned}$$

TRANSFER FUNCTION VIEW - Continued III

Continuing we express C_u, C_y as rational TFs

$$\begin{aligned}C_u &= k^T (sI - A + lc^T)^{-1} b \\&= \frac{k^T \text{adj}(sI - A + lc^T) b}{a_l(s)} \\&= \frac{b_u}{a_l}, \text{ say}\end{aligned}$$

Similarly

$$\begin{aligned}C_y &= k^T (sI - A + lc^T)^{-1} l \\&= \frac{b_y}{a_l}, \text{ say}\end{aligned}$$

NB. Both C_u, C_y are stable transfer functions.
Other 2 DOF representations of LSFB + observer
do not have this important property.

We need naturally to relate b_u, b_y to k, l more
directly.

TRANSFER FUNCTION VIEW - CLOSED LOOP BEHAVIOUR

Referring now to Fig.5.3 loop calculations give

$$\begin{aligned}\bar{y} &= P\bar{u} + P\bar{d} \\ \bar{u} &= \bar{r} - C_u\bar{u} - C_y(\bar{y} + \bar{n}) \\ \Rightarrow &\begin{pmatrix} 1 & -P \\ C_y & 1 + C_u \end{pmatrix} \begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} = \begin{pmatrix} P\bar{d} \\ \bar{r} - C_y\bar{n} \end{pmatrix} \\ \Rightarrow \begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} &= \frac{1}{D} \begin{pmatrix} 1 + C_u & P \\ -C_y & 1 \end{pmatrix} \begin{pmatrix} P\bar{d} \\ \bar{r} - C_y\bar{n} \end{pmatrix} \\ D &= 1 + C_u + PC_y \\ \Rightarrow \bar{y} &= \bar{y}_r + \bar{y}_d + \bar{y}_n \\ &= \frac{P}{D}\bar{r} + \frac{(1 + C_u)P\bar{d}}{D} - \frac{PC_y\bar{n}}{D}\end{aligned}$$

We can read off the CLTF

$$G = CLTF = \frac{P}{D} = \frac{P}{1 + C_u + PC_y}$$

TRANSFER FUNCTION VIEW: POLYNOMIAL INTERPRETATION

We know $G(s) = CLTF = \frac{b(s)}{a_k(s)}$

However from the 2 DOF view

$$\begin{aligned} G(s) &= \frac{P}{1 + C_u + PC_y} \\ &= \frac{\frac{b}{a}}{1 + \frac{b_u}{a_l} + \frac{b}{a} \frac{b_y}{a_l}} \\ &= \frac{ba_l}{a(a_l + b_u) + bb_y} \end{aligned}$$

Note that numerator and denominator each, may have degree up to $2n - 1, 2n$ respectively.

To match this to the CLTF result above we need to show b_u, b_y satisfy

$$a(a_l + b_u) + bb_y = a_l a_k$$

whereupon we will get

$$G(s) = \frac{ba_l}{a_k a_l} = \frac{b}{a_k}$$

NB. Only stable modes are being cancelled.

POLYNOMIAL DESIGN - BEZOUT

We have to solve

$$\begin{aligned}a(a_l + b_u) + bb_y &= a_l a_k \\ \Rightarrow ab_u + bb_y &= (a_k - a)a_l\end{aligned}$$

Note that a_k, a, a_l are monic.

Denote $d_i = a_{ki} - a_i$, $i = 1, \dots, n$ then we find

$$\begin{aligned}& (d_1 s^{n-1} + d_2 s^{n-2} + \dots + d_n)(s^n + a_{l1} s^{n-1} + \dots + a_l) \\ &= (s^n + a_1 s^{n-1} + \dots + a_n)(b_{u1} s^{n-1} + b_{u2} s^{n-2} + \dots + b_u) \\ &+ (b_{y1} s^{n-1} + \dots + b_{yn})(b_1 s^{n-1} + b_2 s^{n-2} + \dots + b_n)\end{aligned}$$

Now match powers of s . The highest power is s^{2n-1} and there are then $2n$ coefficients and so $2n$ equations. We get e.g. with $n = 4$

POLYNOMIAL DESIGN: BEZOUT -

Continued

$$\begin{array}{l}
 {}_s 2n-1 : \\
 {}_s 2n-2 : \\
 \cdot : \\
 \cdot : \\
 \cdot : \\
 \cdot : \\
 \cdot : \\
 \cdot :
 \end{array}
 \begin{pmatrix}
 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 a_1 & 1 & 0 & 0 & b_1 & 0 & 0 & 0 & 0 \\
 a_2 & a_1 & 1 & 0 & b_2 & b_1 & 0 & 0 & 0 \\
 a_3 & a_2 & a_1 & 1 & b_3 & b_2 & b_1 & 0 & 0 \\
 a_4 & a_3 & a_2 & a_1 & b_4 & b_3 & b_2 & b_1 & 0 \\
 0 & a_4 & a_3 & a_2 & 0 & b_4 & b_3 & b_2 & 0 \\
 0 & 0 & a_4 & a_3 & 0 & 0 & b_4 & b_3 & 0 \\
 0 & 0 & 0 & a_4 & 0 & 0 & 0 & b_4 & 0
 \end{pmatrix}
 \begin{pmatrix}
 b_{u1} \\
 b_{u2} \\
 b_{u3} \\
 b_{u4} \\
 b_{y1} \\
 b_{y2} \\
 b_{y3} \\
 b_{y4}
 \end{pmatrix}$$

$$= \begin{pmatrix}
 1 & 0 & 0 & 0 \\
 a_{l1} & 1 & 0 & 0 \\
 a_{l2} & a_{l1} & 1 & 0 \\
 a_{l3} & a_{l2} & a_{l1} & 1 \\
 a_{l4} & a_{l3} & a_{l2} & a_{l1} \\
 0 & a_{l4} & a_{l3} & a_{l2} \\
 0 & 0 & a_{l4} & a_{l3} \\
 0 & 0 & 0 & a_{l4}
 \end{pmatrix}
 \begin{pmatrix}
 d_1 \\
 d_2 \\
 d_3 \\
 d_4
 \end{pmatrix}$$

POLYNOMIAL DESIGN - BEZOUT EQUATION - ContinuedII

or

$$\mathcal{S}(a, b) \begin{pmatrix} b_u \\ b_y \end{pmatrix} = \begin{pmatrix} \mathcal{A}_{l-} \\ \mathcal{A}_{l+} \end{pmatrix} d$$

$$\mathcal{S}(a, b) = \begin{bmatrix} \mathcal{A}_{-} & \mathcal{B}_{-} \\ \mathcal{A}_{+} & \mathcal{B}_{+} \end{bmatrix}$$

$$= \text{Sylvester Matrix}$$

These equations have a unique solution iff $\mathcal{S}(a, b)$ has full rank.

i.e. iff a, b coprime

i.e. iff A, b, c is minimal as expected.

POLYNOMIAL DESIGN - BEZOUT EQUATION -Example

$$P(s) = \frac{1}{s^2} = \frac{0.s + 1}{s^2 + 0.s + 0}$$

Place closed loop control poles at 2nd order system

$$a_k(s) = s^2 + \sqrt{2}s + 1, (\omega_n = 1, \zeta = .7)$$

Place observer poles at 'faster' locations

$$a_l(s) = s^2 + 5s + 25, (\omega_n = 5, \zeta = .5)$$

POLYNOMIAL DESIGN - BEZOUT EQUATION -Example - Continued

Thus

$$\begin{aligned}\Rightarrow \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} &= \begin{pmatrix} a_{k1} - a_1 \\ a_{k2} - a_2 \end{pmatrix} \\ &= \begin{pmatrix} 1.41 - 0 \\ 1 - 0 \end{pmatrix} = \begin{pmatrix} 1.41 \\ 1 \end{pmatrix}\end{aligned}$$

Then

$$\begin{aligned}w &= \begin{pmatrix} \mathcal{A}_{l-} \\ \mathcal{A}_{l+} \end{pmatrix} d \\ &= \begin{pmatrix} 1 & 0 \\ 5 & 1 \\ 25 & 5 \\ 0 & 25 \end{pmatrix} \begin{pmatrix} 1.41 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.41 \\ 8.05 \\ 40.35 \\ 25 \end{pmatrix}\end{aligned}$$

POLYNOMIAL DESIGN: BEZOUT EQUATION -Example -ContinuedII

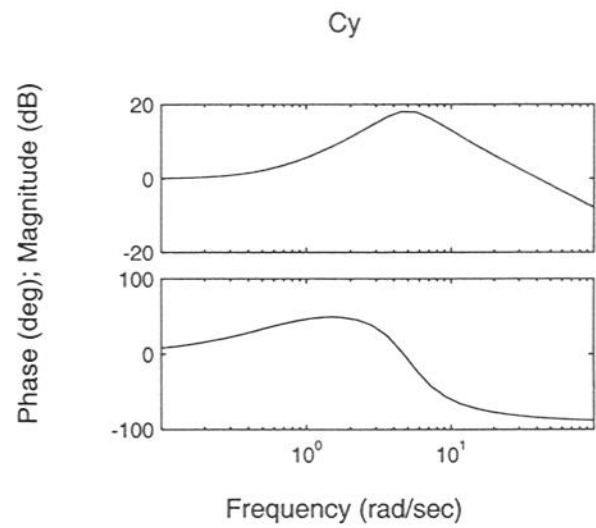
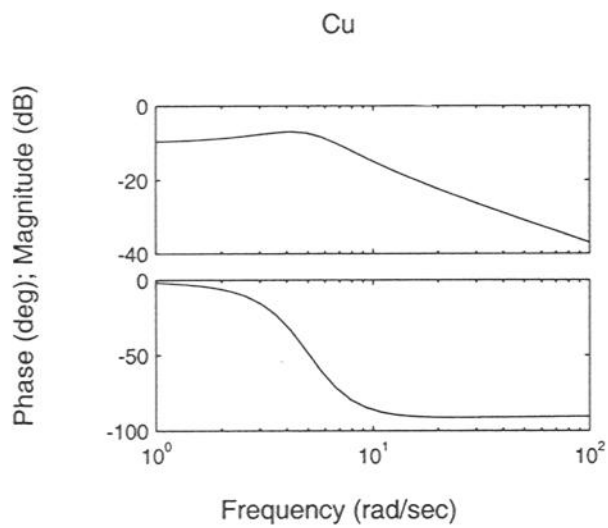
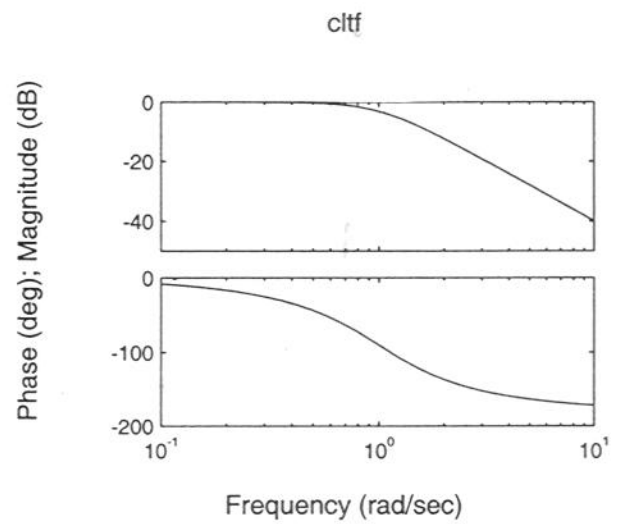
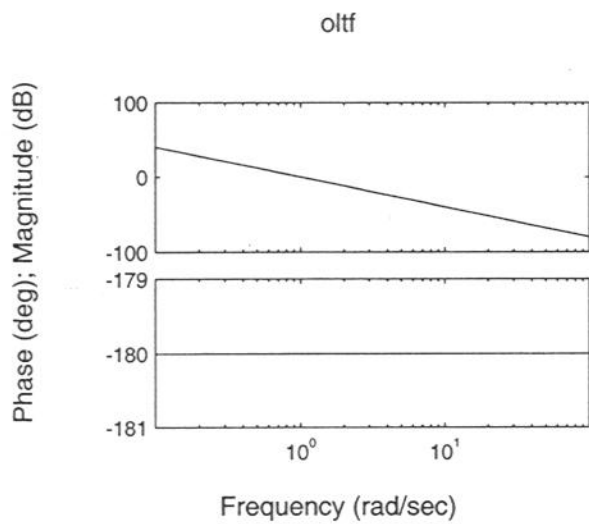
Sylester matrix is

$$\begin{aligned}\mathcal{S}(a, b) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ a_1 & 1 & b_1 & 0 \\ a_2 & a_1 & b_2 & b_1 \\ 0 & a_2 & 0 & b_2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I\end{aligned}$$

So trivially $\mathcal{S}(a, b)$ has full rank, so

$$\begin{aligned}\begin{pmatrix} b_u \\ b_y \end{pmatrix} &= Iw = w \\ \Rightarrow (b_u^{(1)}, b_u^{(2)}) &= (1.42, 8.05) \\ (b_y^{(1)}, b_y^{(2)}) &= (40.35, 25)\end{aligned}$$

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TRANSFER FUNCTION VIEW - SENSITIVITIES

We again read off the CLTF

$$G = CLTF = \frac{P}{D} = \frac{P}{1 + C_u + PC_y}$$

To find the sensitivity function perturb P to find

$$\begin{aligned} dG &= \frac{dP}{D} - \frac{P(C_y dP)}{D^2} \\ \Rightarrow \frac{dG}{G} &= \frac{dP}{P} \left(1 - \frac{PC_y}{D}\right) \\ &= \frac{dP}{P} \frac{1 + C_u}{1 + C_u + PC_y} \\ \Rightarrow \mathcal{S} &= \frac{1 + C_u}{1 + C_u + PC_y} = \frac{1 + C_u}{D} \\ &= \frac{1}{1 + PC_{eff}} \\ C_{eff} &= \frac{C_y}{1 + C_u} \end{aligned}$$

TRANSFER FUNCTION VIEW: SENSITIVITIES- Continued

$$T = 1 - \mathcal{S} = \frac{PC_y}{D} = \frac{PC_y}{1 + C_u + PC_y}$$

Thus

$$CLTF = \frac{P}{D} = \frac{''T''}{C_y}$$

$$NTF = -\frac{PC_y}{D} = -T$$

$$IDTF = \frac{(1 + C_u)P}{D} = \mathcal{S}P$$

So we see, as expected with a 2DOF architecture, independent shaping of NTF and CLTF is possible.

- Tracking error TF is $(\bar{r} - \bar{y} = G_{r \rightarrow e} \bar{r})$

$$G_{r \rightarrow e} = \left(1 - \frac{P}{D}\right)$$

- Disturbance rejection TF is

$$G_{d \rightarrow y} = \frac{(1 + C_u)P}{D} = \mathcal{S}P$$

Sensitivities in TF form

$$\begin{aligned}
 \bullet \mathcal{S} &= \frac{1 + C_u}{D} = \frac{1 + C_u}{1 + C_u + PC_y} \\
 &= \frac{1 + \frac{b_u}{a_l}}{1 + \frac{b_u}{a_l} + \frac{b}{a} \frac{b_y}{a_l}} \\
 &= \frac{a(a_l + b_u)}{a(a_l + b_u) + bb_y} \\
 &= \frac{a(a_l + b_u)}{a_l a_k}
 \end{aligned}$$

$\Rightarrow \mathcal{S}$ is stable

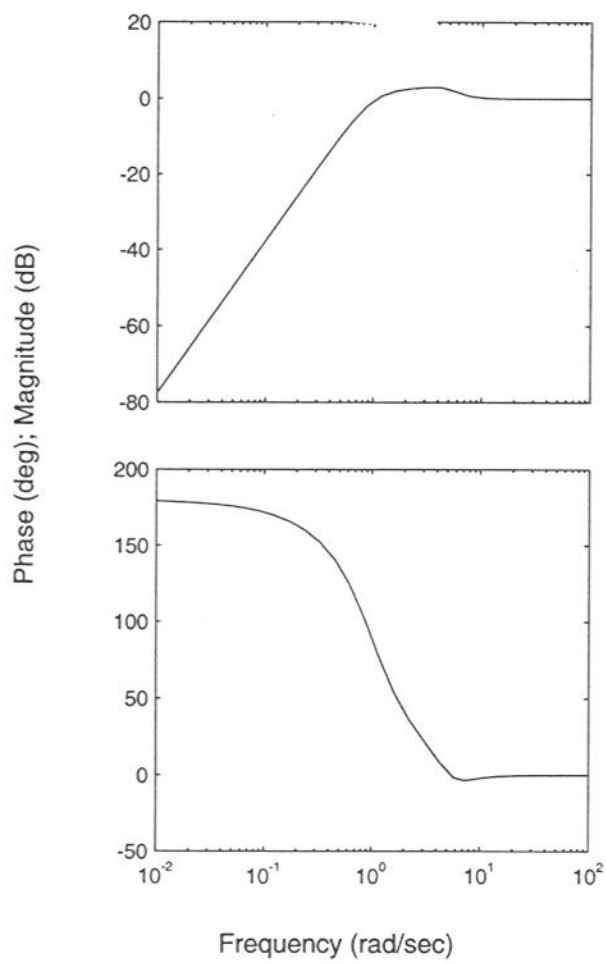
$$\bullet T = 1 - \mathcal{S} = \frac{bb_y}{a_l a_k}$$

by Bezout equation $\Rightarrow T$ is stable

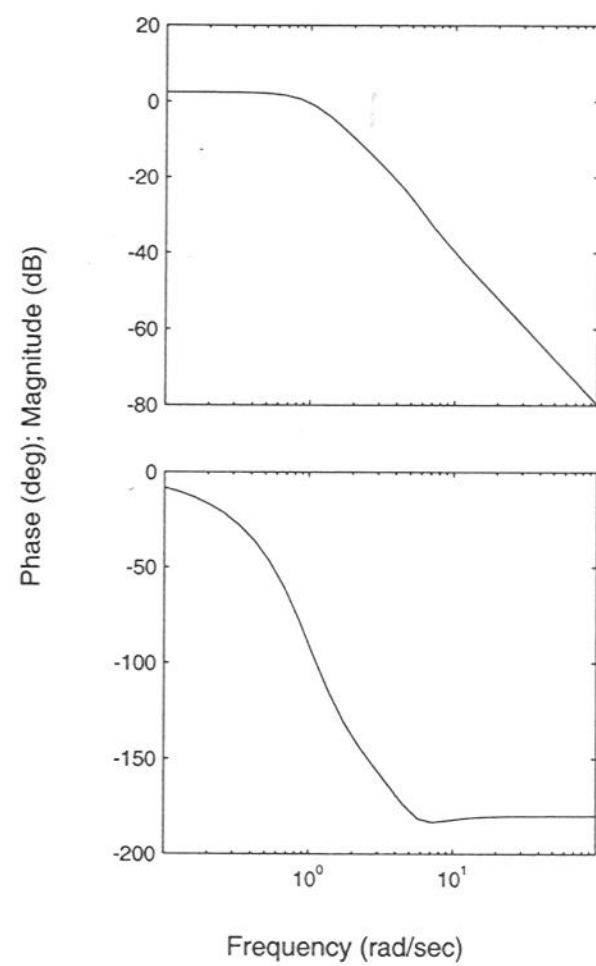
$$\begin{aligned}
 \bullet CLTF &= \frac{b}{a_k}, \text{ stable} \\
 IDTF &= \mathcal{S}P = \left[\frac{a(a_l + b_u)}{a_l a_k} \right] \frac{b}{a} \\
 &= \frac{b(a_l + b_u)}{a_k a_l} \\
 NTF &= -T = -\frac{bb_y}{a_l a_k}, \text{ stable}
 \end{aligned}$$

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sensitivity



IDTF



State Space Connexion - Closing the circle

[Relating b_u, b_y to k, l][Justifying the Bezout equation]

We use the basic MIL relation twice

$$\begin{aligned}(sI - A + lc^T)^{-1} &= (sI - A)^{-1} \\ &\quad - \frac{(sI - A)^{-1}lc^T(sI - A)^{-1}}{1 + c^T(sI - A)^{-1}l}\end{aligned}$$

For future reference, recall

$$\frac{b}{a} = \frac{b(s)}{a(s)} = c^T(sI - A)^{-1}b$$

$$\frac{\bar{l}}{a} = c^T(sI - A)^{-1}l$$

$$\frac{\bar{k}}{a} = k^T(sI - A)^{-1}b$$

$$\text{Define} \quad \frac{\hat{k}}{a} = k^T(sI - A)^{-1}l$$

$$\text{recall} \quad \frac{b_y}{a_l} = k^T(sI - A + lc^T)^{-1}l$$

$$\text{recall} \quad \frac{b_u}{a_l} = k^T(sI - A + lc^T)^{-1}b$$

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State Space Connexion: Closing the circle-

Continued

Using the MIL then, we find

$$\begin{aligned} & k^T (sI - A + lc^T)^{-1} \\ = & k^T (sI - A)^{-1} - \frac{k^T (sI - A)^{-1} lc^T (sI - A)^{-1}}{1 + c^T (sI - A)^{-1} l} \\ = & k^T (sI - A)^{-1} - \frac{\hat{k} c^T (sI - A)^{-1}}{a (1 + \bar{l}/a)} \\ = & k^T (sI - A)^{-1} - \frac{\hat{k}}{a_l} c^T (sI - A)^{-1} \end{aligned}$$

- Now post multiply by l to find

$$\begin{aligned} \frac{b_y}{a_l} &= k^T (sI - A)^{-1} l - \frac{\hat{k}}{a_l} c^T (sI - A)^{-1} l \\ &= \frac{\hat{k}}{a} - \frac{\hat{k}}{a_l} \frac{\bar{l}}{a} \\ &= \frac{\hat{k}}{a} \left(1 - \frac{\bar{l}}{a_l}\right) = \frac{\hat{k}}{a} \frac{(a_l - \bar{l})}{a_l} = \frac{\hat{k}}{a_l} \\ \Rightarrow b_y &= \hat{k} \end{aligned}$$

State Space Connexion: Closing the circle- Continued II

Now post multiply by b to find

$$\begin{aligned}\frac{b_u}{a_l} &= k^T (sI - A)^{-1} b - \frac{b_y}{a_l} c^T (sI - A)^{-1} b \\ &= \frac{\bar{k}}{a} - \frac{b_y}{a_l} \frac{b}{a} \\ \Rightarrow ab_u &= a_l \bar{k} - bb_y \\ \Rightarrow a_l(\bar{k} + a) &= a(b_u + a_l) + bb_y \\ \Rightarrow a_l a_k &= a(b_u + a_l) + bb_y\end{aligned}$$

as required.

State Space Connexion: Closing the circle- Continued III

Can also establish this Bezout relation by evaluating $|sI - A + bk^T + lc^T|$ in two ways. Thus

$$\begin{aligned} & |sI - A + bk^T + lc^T| \\ = & |sI - A + lc^T| |I + (sI - A + lc^T)^{-1}bk^T| \\ = & a_l(1 + k^T(sI - A + lc^T)^{-1}b) \\ = & a_l(1 + \frac{b_u}{a_l}) \\ = & a_l + b_u \end{aligned}$$

Evaluating the determinant in reverse order leads to the Bezout equation (once $\hat{k} = b_y$ is established).

LSFB + Observer SS & Transfer Function Views

Summary

- Design requirements

a, b coprime

or $\equiv A, b, c$ controllable, observable.

- To design the 2DOF , LSFBO controller specify:

poles for observer i.e. a_l

closed loop poles i.e. a_k

Then,

(i)SS design

Compute k, l from Bass-Bura formulae and implement observer and LSFB in SS form.

(ii) TF design

Solve the Bezout/ Sylvester equation for b_u, b_y and implement the 2DOF controller in TF form.